

Mohammed Mahfuz Hossain, Ph.D.

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SENIOR DATA SCIENTIST

Strategic, results-driven data scientist with proven expertise in leveraging advanced analytics and machine learning to extract actionable insights from complex datasets. Accomplished in designing and deploying scalable data solutions for energy, power systems, and enterprise applications. Collaborative technical leader with 12+ published articles in peer-reviewed journals, demonstrated success in directing cross-functional teams, communicating complex technical concepts to non-technical stakeholders, and translating data-driven strategies into measurable business impact. Known for combining technical rigor with strategic problem-solving to deliver innovation across infrastructure modernization, cost optimization, and digital transformation initiatives.

CORE COMPETENCIES

Research | Optimization | Analytics | Algorithm Development | Documentation | Requirements Assessments | Troubleshooting | Quality Assurance | Risk Identification / Mitigation | Plotly | Pandas | Tensorflow | Best Practices | Continuous Process Improvements | Reporting | ETL

TECHNICAL PROFICIENCIES

Computer Languages: Python | R | SQL | MATLAB | LabVIEW | HTML

Software Packages: NLP / LLM | VertexAI | AWS | Tableau | Qlik

CAD Design Packages: PTC CREO Parametric | NX | 3DS Max

EXPERIENCE

TRC Companies, Remote, NY

April 2024 - Present

Senior Consultant (Data Science)

- Developed and implemented an automated data collection process in collaboration with Cloud Services team, leveraging AWS and Azure backends to support large-scale pilots
- Engineered scalable solutions enabling major initiatives including CEC EPIC electrification demos and NYSEERDA HELPS programs with robust data collection, storage, and scripting capabilities
- Demonstrated persistent technical leadership over 12+ months, resulting in a fully operational multi-cloud infrastructure solution
- Automated reach code cost effectiveness analysis process using Python scripts, eliminating manual interventions and significantly reducing time and error potential
- Developed professional output workflows enabling efficient iteration and scalability across cost analysis projects
- Created comprehensive documentation supporting long-term maintainability and team adoption of automated analysis tools
- Transformed customer acquisition strategy by implementing data-driven identification methods, shifting from broad outreach to precision targeting and improving market penetration efficiency.
- Engineered a scalable API-driven system for automated data downloads, minimizing manual intervention and freeing up resources for higher-value work.

GHD, INC., Buffalo, NY

June 2019 – March 2024

Senior Data Scientist

Supervised a team in the Agile development of cutting-edge Oil / Gas pipeline optimization digital twin software.

- Delivered quantifiable impact through AI/ML pipeline optimization models, achieving approximately 10% reduction in operating costs (\$160,000 in monthly savings) for major oil and gas infrastructure.
- Collaborated with cross-functional teams to understand business needs and solve complex problems.
- Engineered AI / ML predictive models for Oil/ Gas pipeline components using Python including Pandas, Numpy, Scikit-Learn, Keras, etc. to build the digital twin.
- Collaborated with a cross-functional team to develop the digital twin and solve business problems.
- Presented optimized results to clients through comprehensive reports and visualizations.
- Ensured a deep understanding of client's business needs to tailor solutions effectively.
- Conducted exploratory statistical data analysis with an engineering focus to detect and elucidate anomalies in operational data.

CLARKSON UNIVERSITY, Potsdam, NY**January 2016 - May 2019****Research Assistant, Electrical Engineering**

Conducted advanced research in human-computer interaction, power systems analytics, and machine learning applications for grid resilience and situational awareness.

- Conducted survey research to determine HMI display preferences.
- Authored and presented research findings at the IEEE conference.
- Created interactive visualizations using the R Shiny package for power systems utility outage data.
- Applied ML Models for electricity load forecasting.
- Contributed to a microgrid research project aimed at bolstering the resilience of the electric grid.
- Enhanced situation awareness for control room operators through power system visualizations.

UNIVERSITY OF MALAYA, Malaysia**January 2012 - February 2014****Research assistant, Mechanical Engineering**

Conducted advanced research in thermal energy storage, renewable energy systems, and energy efficiency optimization.

- Authored numerous articles published in reputable journals.
- Presented research findings in International Conference on Automation, Mechanics, and Robotics.
- Conducted experimental setups for a double tube heat exchanger in thermal energy storage applications.
- Executed comprehensive energy, exergy, and cost analyses for renewable energy sources.

BOSCH Automotive and Spares, Abedin Equipment Ltd., Bangladesh
Engineer (Supply Chain)**January 2011 – December 2011**

- Utilized Enterprise Resource Planning (ERP) software to identify and manage large deadstock inventory.
- Conducted market visits to assess the situation of deadstock products and researched alternative products, resulting in a recovery of approximately \$15,000.
- Fostered strong supplier / customer relations through frequent communication, reducing lead time from 4 months to 2 months.

DIRD COMPOSITE TEXTILES LTD., Bangladesh
Engineer (Supply Chain)**January 2010 - December 2010**

- Directed a team of over 100 employees.
- Managed end-to-end operations of the store department, including receiving raw materials/yarn and shipping.

VOLUNTEER WORK**Elected Vice President** of Muslim Students Association (MSA) at Clarkson University January 2016 - December 2017

- Successfully organized a fundraising event for refugees.
- Orchestrated culturally enriching events with themes such as "Muslims Around the World," "Eid Celebration," and "World Hijab Day," attracting hundreds of participants.

Tutoring (Science & Math), Darul Uloom New York, NY

March 2014 – December 2015

- Instructed students across multiple grade levels.
- Employed creative strategies to facilitate students' comprehension of mathematical concepts.

EDUCATION**Doctor of Philosophy (Ph.D.) in Engineering Science**
Clarkson University, Potsdam, NY

Jul 2024

Master of Science (M.S.) in Data Analytics
Clarkson University, Potsdam, NY

May 2019

Master of Science (M.S.) in Engineering Science (Research)
University of Malaya, Malaysia

Jan 2014

Bachelor of Science (B.S.) in Industrial Engineering
SUST, Bangladesh

Aug 2009

PUBLICATIONS

More than twelve articles have been published in international scientific journals. Google Scholar ID: [M.H. Mahfuz](#)

- M. M. Hossain, T. Ortmeyer, and E. Hall, "Human-Centered Design, Development, and Evaluation of an Interface for a Microgrid Controller," *Advances in Science, Technology and Engineering Systems Journal*, vol. 8, no. 3, pp. 80–88, 2023.
- M. H. Mahfuz, T. Ortmeyer, and E. Hall, "Development of a Microgrid Controller Interface Using Human-Centered Design Approach," in *IEEE Power and Energy Society General Meeting*, 2022, vol. 2022-July.
- M. H. Mahfuz, M. R. Anisur, M. a. Kibria, R. Saidur, and I. H. S. C. Metselaar, "Performance investigation of thermal energy storage system with Phase Change Material (PCM) for solar water heating application," *International Communications in Heat and Mass Transfer*, vol. 57, pp. 132–139, Oct. 2014.
- M. H. Mahfuz et al., "Exergetic analysis of a solar thermal power system with PCM storage," *Energy Conversion and Management*, vol. 78, pp. 486–492, Feb. 2014.
- M. H. Mahfuz, M. R. Anisur, M. A. Kibria, I. Metselaar, and R. Saidur, "Performance and cost analysis of building heating system with phase change materials," in *2nd International conference on machine learning, electrical, electronics & mechanical engineering*, 2013, pp. 1–3.